Trainings

General Information

Perform maintenance at whichever interval occurs first. At each scheduled maintenance interval, perform all previous maintenance checks that are required for scheduled maintenance.

Cummins Inc. has found that engines in most applications will **not** experience significant valve/injector train wear. It is recommended that valve and injector adjustment is performed **only** when injectors are removed or when other repairs disturb the valve train.

The QSK45 and QSK60 Series engines have the following lubricating oil filtration options:

Marine and Industrial

- Spin-on Oil Filter without Centrifuge and without Centinel™
- Spin-on Oil Filter with Centrifuge and without Centinel™
- Spin-on Oil Filter without Centrifuge and with Centinel™
- Spin-on Oil Filter with Centrifuge and with Centinel™
- Eliminator™ without Centinel™
- Eliminator™ with Centinel™.

Power Generation

- Spin-on Oil Filter without Centrifuge and without Centinel™
- Spin-on Oil Filter with Centrifuge and without Centinel™
- Spin-on Oil Filter without Centrifuge and with Centinel™
- Spin-on Oil Filter with Centrifuge and with Centinel™
- Eliminator™ without Centinel™
- Eliminator™ with Centinel™.

See the applicable maintenance schedule for the specific oil filtration option used.

Marine and Industrial

Spin-on Oil Filter without Centrifuge and without Centinel™

Maintenance Procedures at Daily Interval

- Pre-lubricating Oil Pump - Check
- Air Cleaner Precleaner - Clean

- Air Cleaner Restriction - Check
- Coolant Level - Check
- Lubricating Oil Level - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Fuel Water Separator - Drain
- Air Tanks and Reservoirs - Drain
- Intake Air Piping - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 Months

- Air Compressor Air Cleaner Element - Check
- Air Leaks, Compressed Air System - Check
- Drive Belt Alternator - Measure
- Fuel Filter (Spin-On Type)[1, 4] - Change
- Fuel Filter (Stage 1)[2, 4] - Change
- Fuel Filter (Stage 2)[2, 4] - Change
- Drive Belt - Check
- Cooling Fan - Check
- Belt Tensioner, Automatic (Alternator) - Check
- Coolant Filter - Change[4]
- Lubricating Oil and Filters - Change[4, 5]
- Oil Level Sensor - Clean and Check
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) - Check[4]
- Hoses, Engine - Check
- Zinc Anode - Check
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- CENSE™ Datalogger - Reset (CM530 **only**. CM2330 does **not** need to be reset.)
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1,000 Hours or 1 Year

- Pre-Lubricating Oil Pump - Check

Maintenance Procedures at 10,000 Hours or 2 Years

- Engine - Clean
- Fan Drive Idler Arm Assembly - Check
- Fan Drive Idler Pulley Assembly - Adjust
- Batteries - Check

- Cold Weather Starting Aids - Check
- Cooling System - Flush[4]
- Fan Hub, Belt Driven - Check
- Water Pump - Check
- Turbocharger - Check (All Applications **EXCEPT** Single Stage Turbocharged Haul Truck Applications operating above 4000 feet altitude)
- Air Compressor Unloader and Valve Assembly - Check
- Vibration Damper, Viscous - Check

Maintenance Procedures at 10,000 Hours

- Turbocharger - Replace (Single Stage Turbocharged Haul Truck Applications operating above 4000 feet altitude)
- Fuel Injectors - Replace[3]

Spin-on Oil Filter with Centrifuge and without Centinel™

Maintenance Procedures at Daily Interval

- Pre-lubricating Oil Pump - Check
- Air Cleaner Precleaner - Clean
- Air Cleaner Restriction - Check
- Coolant Level - Check
- Lubricating Oil Level - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Fuel Water Separator - Drain
- Air Tanks and Reservoirs - Drain
- Intake Air Piping - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 months

- Air Compressor Air Cleaner Element - Check
- Air Leaks, Compressed Air System - Check
- Drive Belt Alternator - Measure
- Fuel Filter (Spin-On Type)[1, 4] - Change
- Fuel Filter (Stage 1)[2, 4] - Change
- Fuel Filter (Stage 2)[2, 4] - Change
- Drive Belt - Check
- Cooling Fan - Check
- Belt Tensioner, Automatic (Alternator) - Check
- Coolant Filter - Change[4]
- Lubricating Oil and Filters- Change[4, 5]
- Oil Level Sensor - Clean and Check

- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) - Check[4]
- Hoses, Engine - Check
- Zinc Anode - Check
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- CENSE™ Datalogger - Reset (CM530 **only**. CM2330 does **not** need to be reset.)
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1,000 Hours or 1 Year

- Pre-Lubricating Oil Pump - Check

Maintenance Procedures at 1500 Hours or 1 Year

- Fleetguard® Centrifuge - Clean and Check[4]

Maintenance Procedures at 10,000 Hours

- Engine - Clean
- Fan Drive Idler Arm Assembly - Check
- Fan Drive Idler Pulley Assembly - Adjust
- Batteries - Check
- Cold Weather Starting Aids - Check
- Cooling System - Flush[4]
- Fan Hub, Belt Driven - Check
- Water Pump - Check
- Turbocharger - Check (All Applications **EXCEPT** Single Stage Turbocharged Haul Truck Applications operating above 4000 feet altitude)
- Air Compressor Unloader and Valve Assembly - Check
- Vibration Damper, Viscous - Check

Maintenance Procedures at 10,000 Hours

- Turbocharger - Replace (Single Stage Turbocharged Haul Truck Applications operating above 4000 feet altitude)
- Fuel Injectors - Replace[3]

Spin-on Oil Filter without Centrifuge and with Centinel™

Maintenance Procedures at Daily Interval

- Pre-lubricating Oil Pump - Check
- Air Cleaner Precleaner - Clean
- Air Cleaner Restriction - Check
- Coolant Level - Check
- Lubricating Oil Level - Check

- Centinel™ Oil Level - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Fuel Water Separator - Drain
- Air Tanks and Reservoirs - Drain
- Intake Air Piping - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 Months

- Air Compressor Air Cleaner Element - Check
- Air Leaks, Compressed Air System - Check
- Drive Belt Alternator - Measure
- Fuel Filter (Spin-On Type)[1, 4] - Change
- Fuel Filter (Stage 1)[2, 4] - Change
- Fuel Filter (Stage 2)[2, 4] - Change
- Drive Belts - Check
- Cooling Fan - Check
- Oil Analysis (reference the following bulletins):
    - Fluids for Cummins® Products Service Manual, Bulletin 5411406, Section 4 - Engine Oil
    - Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060. (/qs3/pubsys2/xml/en/bulletin/4022060.html)
    - Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452. (/qs3/pubsys2/xml/en/bulletin/2883452.html)
- Belt Tensioner, Automatic (Alternator) - Check
- Coolant Filter - Change[4]
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) - Check[4]
- Hoses, Engine - Check
- Zinc Anode - Check
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- CENSE™ Datalogger - Reset (CM530 **only**. CM2330 does **not** need to be reset.)
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1000 Hours or 1 Year

- Lubricating Oil Filters - Change[4, 5]
- Oil Level Sensor - Clean and Check
- Pre-Lubricating Oil Pump - Check

Maintenance Procedures at 10,000 Hours or 2 Years

- Engine - Clean
- Fan Drive Idler Arm Assembly - Check
- Fan Drive Idler Pulley Assembly - Adjust
- Batteries - Check
- Cold Weather Starting Aids - Check
- Cooling System - Flush[4]
- Fan Hub, Belt Driven - Check
- Water Pump - Check
- Turbocharger - Check (All Applications **EXCEPT** Single Stage Turbocharged Haul Truck Applications operating above 4000 feet altitude)
- Air Compressor Unloader and Valve Assembly - Check
- Vibration Damper, Viscous - Check

Maintenance Procedures at 10,000 Hours

- Turbocharger - Replace (Single Stage Turbocharged Haul Truck Applications operating above 4000 feet altitude)
- Fuel Injectors - Replace[3]

Spin-on Oil Filter with Centrifuge and with Centinel™

Maintenance Procedures at Daily Interval

- Pre-lubricating Oil Pump - Check
- Air Cleaner Precleaner - Clean
- Air Cleaner Restriction - Check
- Coolant Level - Check
- Lubricating Oil Level - Check
- Centinel™ Oil Level - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Fuel Water Separator - Drain
- Air Tanks and Reservoirs - Drain
- Intake Air Piping - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 hours or 6 Months

- Air Compressor Air Cleaner Element - Check
- Air Leaks, Compressed Air System - Check
- Drive Belt Alternator - Measure
- Fuel Filter (Spin-On Type)[1, 4] - Change
- Fuel Filter (Stage 1)[2, 4] - Change
- Fuel Filter (Stage 2)[2, 4] - Change
- Drive Belts - Check

- Oil Analysis (reference the following bulletins):
  - Fluids for Cummins® Products Service Manual, Bulletin 5411406, Section 4 - Engine Oil.
  - Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060. (/qs3/pubsys2/xml/en/bulletin/4022060.html)
  - Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452. (/qs3/pubsys2/xml/en/bulletin/2883452.html)
- Cooling Fan - Check
- Belt Tensioner, Automatic (Alternator) - Check
- Coolant Filter - Change[4]
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) - Check[4]
- Hoses, Engine - Check
- Zinc Anode - Check
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- CENSE™ Datalogger - Reset (CM530 **only**. CM2330 does **not** need to be reset.)
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1000 Hours or 1 Year

- Lubricating Oil Filters - Change[4, 5]
- Oil Level Sensor - Clean and Check
- Pre-Lubricating Oil Pump - Check

Maintenance Procedures at 1500 Hours or 1 Year

- Fleetguard® Centrifuge - Clean and Check[4]

Maintenance Procedures at 10,000 Hours or 2 Years

- Engine - Clean
- Fan Drive Idler Arm Assembly - Check
- Fan Drive Idler Pulley Assembly - Adjust
- Batteries - Check
- Cold Weather Starting Aids - Check
- Cooling System - Flush[4]
- Fan Hub, Belt Driven - Check
- Water Pump - Check
- Turbocharger - Check (All Applications **EXCEPT** Single Stage Turbocharged Haul Truck Applications operating above 4000 feet altitude)
- Air Compressor Unloader and Valve Assembly - Check
- Vibration Damper, Viscous - Check

Maintenance Procedures at 10,000 Hours

- Turbocharger - Replace (Single Stage Turbocharged Haul Truck Applications operating above 4000 feet altitude)
- Fuel Injectors - Replace[3]

Eliminator™ without Centinel™

Maintenance Procedures at Daily Interval

- Pre-lubricating Oil Pump - Check
- Air Cleaner Precleaner - Clean
- Air Cleaner Restriction - Check
- Coolant Level - Check
- Lubricating Oil Level - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Fuel Water Separator - Drain
- Air Tanks and Reservoirs - Drain
- Eliminator™ Filter Rotation - Check
- Intake Air Piping - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 Months

- Air Shutoff Valve - Check
- Air Compressor Air Cleaner Element - Check
- Air Leaks, Compressed Air System - Check
- Drive Belt Alternator - Measure
- Fuel Filter (Spin-On Type)[1, 4] - Change
- Fuel Filter (Stage 1)[2, 4] - Change
- Fuel Filter (Stage 2)[2, 4] - Change
- Drive Belts - Check
- Cooling Fan - Check
- Belt Tensioner, Automatic (Alternator) - Check
- Coolant Filter - Change[4]
- Lubricating Oil - Change[4, 5]
- Oil Level Sensor - Clean and Check
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) - Check[4]
- Hoses, Engine - Check
- Zinc Anode - Check
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- CENSE™ Datalogger - Reset (CM530 **only**. CM2330 does **not** need to be reset.)
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1000 Hours or 1 Year

- Eliminator™ Filter Centrifuge - Clean and Check[4]
- Eliminator™ Filter Pressure - Check
- Pre-Lubricating Oil Pump - Check

Maintenance Procedures at 10,000 Hours or 2 Years

- Engine - Clean
- Fan Drive Idler Arm Assembly - Check
- Fan Drive Idler Pulley Assembly - Adjust
- Batteries - Check
- Cold Weather Starting Aids - Check
- Cooling System - Flush[4]
- Fan Hub, Belt Driven - Check
- Water Pump - Check
- Turbocharger - Check (All Applications **EXCEPT** Single Stage Turbocharged Haul Truck Applications operating above 4000 feet altitude)
- Air Compressor Unloader and Valve Assembly - Check
- Vibration Damper, Viscous - Check

Maintenance Procedures at 10,000 Hours

- Turbocharger - Replace (Single Stage Turbocharged Haul Truck Applications operating above 4000 feet altitude)
- Fuel Injectors - Replace[3]

Eliminator™ with Centinel™

If application is **not** specified in the title, assume all engines are affected.

Maintenance Procedures at Daily Interval

- Pre-lubricating Oil Pump - Check
- Air Cleaner Precleaner - Clean
- Air Cleaner Restriction - Check
- Coolant Level - Check
- Lubricating Oil Level - Check
- Centinel™ Oil Level - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Fuel Water Separator - Drain
- Air Tanks and Reservoirs - Drain
- Eliminator™ Filter Rotation - Check
- Intake Air Piping - Maintenance
- Fuel Filter, Remote-Mounted[2] - Drain

- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 Months

- Air Shutoff Valve - Check
- Air Compressor Air Cleaner Element - Check
- Air Leaks, Compressed Air System - Check
- Drive Belt Alternator - Measure
- Fuel Filter (Spin-On Type)[1, 4] - Change
- Fuel Filter (Stage 1)[2, 4] - Change
- Fuel Filter (Stage 2)[2, 4] - Change
- Drive Belts - Check
- Cooling Fan - Check
- Oil Analysis (reference the following bulletins):
  - Fluids for Cummins® Products Service Manual, Bulletin 5411406, Section 4 - Engine Oil.
  - Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060. (/qs3/pubsys2/xml/en/bulletin/4022060.html)
  - Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452. (/qs3/pubsys2/xml/en/bulletin/2883452.html)
- Belt Tensioner, Automatic (Alternator) - Check
- Coolant Filter - Change[4]
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) - Check[4]
- Hoses, Engine - Check
- Zinc Anode - Check
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- CENSE™ Datalogger - Reset (CM530 **only**. CM2330 does **not** need to be reset.)
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1000 Hours or 1 Year

- Eliminator™ Filter Centrifuge - Clean and Check[4]
- Oil Level Sensor - Clean and Check
- Eliminator™ Filter Pressure - Check
- Pre-Lubricating Oil Pump - Check

Maintenance Procedures at 10,000 Hours or 2 Years

- Engine - Clean
- Fan Drive Idler Arm Assembly - Check
- Fan Drive Idler Pulley Assembly - Adjust
- Batteries - Check

- Cold Weather Starting Aids - Check
- Cooling System - Flush[4]
- Fan Hub, Belt Driven - Check
- Water Pump - Check
- Turbocharger - Check (All Applications **EXCEPT** Single Stage Turbocharged Haul Truck Applications operating above 4000 feet altitude)
- Air Compressor Unloader and Valve Assembly - Check
- Vibration Damper, Viscous - Check

Maintenance Procedures at 10,000 Hours

- Turbocharger - Replace (Single Stage Turbocharged Haul Truck Applications operating above 4000 feet altitude)
- Fuel Injectors - Replace[3]

Power Generation

Spin-on Oil Filter without Centrifuge and without Centinel™

Maintenance Procedures at Daily Interval

- Pre-lubricating Oil Pump - Check
- Air Cleaner Precleaner - Clean
- Air Cleaner Restriction - Check
- Coolant Level - Check
- Lubricating Oil Level - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Fuel Water Separator - Drain
- Air Tanks and Reservoirs - Drain
- Intake Air Piping - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 Months

- Air Compressor Air Cleaner Element - Check
- Air Leaks, Compressed Air System - Check
- Drive Belt Alternator - Measure
- Fuel Filter (Spin-On Type)[1, 4] - Change
- Fuel Filter (Stage 1)[2, 4] - Change
- Fuel Filter (Stage 2)[2, 4] - Change
- Drive Belt - Check
- Cooling Fan - Check

- Belt Tensioner, Automatic (Alternator) - Check
- Coolant Filter - Change[4]
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) - Check[4]
- Hoses, Engine - Check
- Zinc Anode - Check
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- CENSE™ Datalogger - Reset (CM530 **only**. CM2330 does **not** need to be reset.)
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 250 Hours or 1 Year

- Lubricating Oil and Filters - Change[4, 5]
- Oil Level Sensor - Clean and Check

Maintenance Procedures at 10,000 Hours or 2 Years

- Engine - Clean
- Fan Drive Idler Arm Assembly - Check
- Fan Drive Idler Pulley Assembly - Adjust
- Batteries - Check
- Cold Weather Starting Aids - Check
- Cooling System - Flush[4]
- Fan Hub, Belt Driven - Check
- Water Pump - Check
- Turbocharger - Check
- Air Compressor Unloader and Valve Assembly - Check
- Vibration Damper, Viscous - Check

Maintenance Procedures at 10,000 Hours

- Fuel Injectors - Replace[3]

Spin-on Oil Filter with Centrifuge and without Centinel™

Maintenance Procedures at Daily Interval

- Pre-lubricating Oil Pump - Check
- Air Cleaner Precleaner - Clean
- Air Cleaner Restriction - Check
- Coolant Level - Check
- Lubricating Oil Level - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Fuel Water Separator - Drain

- Air Tanks and Reservoirs - Drain
- Intake Air Piping - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 months

- Air Compressor Air Cleaner Element - Check
- Air Leaks, Compressed Air System - Check
- Drive Belt Alternator - Measure
- Fuel Filter (Spin-On Type)[1, 4] - Change
- Fuel Filter (Stage 1)[2, 4] - Change
- Fuel Filter (Stage 2)[2, 4] - Change
- Drive Belt - Check
- Cooling Fan - Check
- Belt Tensioner, Automatic (Alternator) - Check
- Coolant Filter - Change[4]
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) - Check[4]
- Hoses, Engine - Check
- Zinc Anode - Check
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- CENSE™ Datalogger - Reset (CM530 **only**. CM2330 does **not** need to be reset.)
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 250 Hours or 1 Year

- Lubricating Oil and Filters - Change[4, 5]
- Lubricating Oil Level - Check
- Oil Level Sensor - Clean and Check

Maintenance Procedures at 1500 Hours or 1 Year

- Fleetguard® Centrifuge - Clean and Check[4]

Maintenance Procedures at 10,000 Hours or 2 Years

- Engine - Clean
- Fan Drive Idler Arm Assembly - Check
- Fan Drive Idler Pulley Assembly - Adjust
- Batteries - Check
- Cold Weather Starting Aids - Check
- Cooling System - Flush[4]

- Fan Hub, Belt Driven - Check
- Water Pump - Check
- Turbocharger - Check
- Air Compressor Unloader and Valve Assembly - Check
- Vibration Damper, Viscous - Check

Maintenance Procedures at 10,000 Hours

- Fuel Injectors - Replace[3]

Spin-on Oil Filter without Centrifuge and with Centinel™

Maintenance Procedures at Daily Interval

- Pre-lubricating Oil Pump - Check
- Air Cleaner Precleaner - Clean
- Air Cleaner Restriction - Check
- Coolant Level - Check
- Lubricating Oil Level - Check
- Centinel™ Oil Level - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Fuel Water Separator - Drain
- Air Tanks and Reservoirs - Drain
- Intake Air Piping - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 Months

- Air Compressor Air Cleaner Element - Check
- Air Leaks, Compressed Air System - Check
- Drive Belt Alternator - Measure
- Fuel Filter (Spin-On Type)[1, 4] - Change
- Fuel Filter (Stage 1)[2, 4] - Change
- Fuel Filter (Stage 2)[2, 4] - Change
- Drive Belts - Check
- Cooling Fan - Check
- Oil Analysis (reference the following bulletins):
  - Fluids for Cummins® Products Service Manual, Bulletin 5411406, Section 4 - Engine Oil.
  - Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060. (/qs3/pubsys2/xml/en/bulletin/4022060.html)
  - Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452. (/qs3/pubsys2/xml/en/bulletin/2883452.html)
- Belt Tensioner, Automatic (Alternator) - Check

- Coolant Filter - Change[4]
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) - Check[4]
- Hoses, Engine - Check
- Zinc Anode - Check
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- CENSE™ Datalogger - Reset (CM530 **only**. CM2330 does **not** need to be reset.)
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1000 Hours or 1 Year

- Lubricating Oil Filters - Change[4, 5]
- Oil Level Sensor - Clean and Check

Maintenance Procedures at 10,000 Hours or 2 Years

- Engine - Clean
- Fan Drive Idler Arm Assembly - Check
- Fan Drive Idler Pulley Assembly - Adjust
- Batteries - Check
- Cold Weather Starting Aids - Check
- Cooling System - Flush[4]
- Fan Hub, Belt Driven - Check
- Water Pump - Check
- Turbocharger - Check
- Air Compressor Unloader and Valve Assembly - Check
- Vibration Damper, Viscous - Check

Maintenance Procedures at 10,000 Hours

- Fuel Injectors - Replace[3]

Spin-on Oil Filter with Centrifuge and with Centinel™

Maintenance Procedures at Daily Interval

- Pre-lubricating Oil Pump - Check
- Air Cleaner Precleaner - Clean
- Air Cleaner Restriction - Check
- Coolant Level - Check
- Lubricating Oil Level - Check
- Centinel™ Oil Level - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Fuel Water Separator - Drain

- Air Tanks and Reservoirs - Drain
- Intake Air Piping - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 hours or 6 Months

- Air Compressor Air Cleaner Element - Check
- Air Leaks, Compressed Air System - Check
- Drive Belt Alternator - Measure
- Fuel Filter (Spin-On Type)[1, 4] - Change
- Fuel Filter (Stage 1)[2, 4] - Change
- Fuel Filter (Stage 2)[2, 4] - Change
- Drive Belts - Check
- Cooling Fan - Check
- Oil Analysis (reference the following bulletins):
    - Fluids for Cummins® Products Service Manual, Bulletin 5411406, Section 4 - Engine Oil.
    - Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060. (/qs3/pubsys2/xml/en/bulletin/4022060.html)
    - Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452. (/qs3/pubsys2/xml/en/bulletin/2883452.html)
- Belt Tensioner, Automatic (Alternator) - Check
- Coolant Filter - Change[4]
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) - Check[4]
- Hoses, Engine - Check
- Zinc Anode - Check
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- CENSE™ Datalogger - Reset (CM530 **only**. CM2330 does **not** need to be reset.)
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1000 Hours or 1 Year

- Lubricating Oil Filters - Change[4, 5]
- Oil Level Sensor - Clean and Check

Maintenance Procedures at 1500 Hours or 1 Year

- Fleetguard® Centrifuge - Clean and Check[4]

Maintenance Procedures at 10,000 Hours or 2 Years

- Engine - Clean
- Fan Drive Idler Arm Assembly - Check
- Fan Drive Idler Pulley Assembly - Adjust
- Batteries - Check
- Cold Weather Starting Aids - Check
- Cooling System - Flush[4]
- Fan Hub, Belt Driven - Check
- Water Pump - Check
- Turbocharger - Check
- Air Compressor Unloader and Valve Assembly - Check
- Vibration Damper, Viscous - Check

Maintenance Procedures at 10,000 Hours

- Fuel Injectors - Replace[3]

Eliminator™ without Centinel™

Maintenance Procedures at Daily Interval

- Pre-lubricating Oil Pump - Check
- Air Cleaner Precleaner - Clean
- Air Cleaner Restriction - Check
- Coolant Level - Check
- Lubricating Oil Level - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Fuel Water Separator - Drain
- Air Tanks and Reservoirs - Drain
- Eliminator Filter Rotation - Check
- Intake Air Piping - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 Months

- Air Compressor Air Cleaner Element - Check
- Air Leaks, Compressed Air System - Check
- Drive Belt Alternator - Measure
- Fuel Filter (Spin-On Type)[1, 4] - Change
- Fuel Filter (Stage 1)[2, 4] - Change
- Fuel Filter (Stage 2)[2, 4] - Change
- Drive Belts - Check
- Cooling Fan - Check
- Belt Tensioner, Automatic (Alternator) - Check
- Coolant Filter - Change[4]

- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) - Check[4]
- Hoses, Engine - Check
- Zinc Anode - Check
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- CENSE™ Datalogger - Reset (CM530 **only**. CM2330 does **not** need to be reset.)
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 250 Hours or 1 Year

- Lubricating Oil - Change[4, 5]
- Oil Level Sensor - Clean and Check

Maintenance Procedures at 1000 Hours or 1 Year

- Eliminator™ Filter Centrifuge - Clean and Check[4]
- Eliminator™ Filter Pressure - Check

Maintenance Procedures at 10,000 Hours or 2 Years

- Engine - Clean
- Fan Drive Idler Arm Assembly - Check
- Fan Drive Idler Pulley Assembly - Adjust
- Batteries - Check
- Cold Weather Starting Aids - Check
- Cooling System - Flush[4]
- Fan Hub, Belt Driven - Check
- Water Pump - Check
- Turbocharger - Check
- Air Compressor Unloader and Valve Assembly - Check
- Vibration Damper, Viscous - Check

Maintenance Procedures at 10,000 Hours

- Fuel Injectors - Replace[3]

Eliminator™ with Centinel™

Maintenance Procedures at Daily Interval

- Pre-lubricating Oil Pump - Check
- Air Cleaner Precleaner - Clean
- Air Cleaner Restriction - Check
- Coolant Level - Check
- Lubricating Oil Level - Check

- Centinel™ Oil Level - Check
- Marine Gear - Check
- Sea Water Strainer - Clean
- Fuel Water Separator - Drain
- Air Tanks and Reservoirs - Drain
- Eliminator™ Filter Rotation - Check
- Intake Air Piping - Maintenance
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 Months

- Air Compressor Air Cleaner Element - Check
- Air Leaks, Compressed Air System - Check
- Drive Belt Alternator - Measure
- Fuel Filter (Spin-On Type)[1, 4] - Change
- Fuel Filter (Stage 1)[2, 4] - Change
- Fuel Filter (Stage 2)[2, 4] - Change
- Drive Belts - Check
- Cooling Fan - Check
- Oil Analysis (reference the following bulletins):
    - Fluids for Cummins® Products Service Manual, Bulletin 5411406, Section 4 - Engine Oil.
    - Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060. (/qs3/pubsys2/xml/en/bulletin/4022060.html)
    - Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452. (/qs3/pubsys2/xml/en/bulletin/2883452.html)
- Belt Tensioner, Automatic (Alternator) - Check
- Coolant Filter - Change[4]
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) - Check[4]
- Hoses, Engine - Check
- Zinc Anode - Check
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- CENSE™ Datalogger - Reset (CM530 **only**. CM2330 does **not** need to be reset.)
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1000 Hours or 1 Year

- Eliminator™ Filter Centrifuge - Clean and Check[4]
- Oil Level Sensor - Clean and Check
- Eliminator™ Filter Pressure - Check

Maintenance Procedures at 10,000 Hours or 2 Years

- Engine - Clean
- Fan Drive Idler Arm Assembly - Check
- Fan Drive Idler Pulley Assembly - Adjust
- Batteries - Check
- Cold Weather Starting Aids - Check
- Cooling System - Flush[4]
- Fan Hub, Belt Driven - Check
- Water Pump - Check
- Turbocharger - Check
- Air Compressor Unloader and Valve Assembly - Check
- Vibration Damper, Viscous - Check

Maintenance Procedures at 10,000 Hours

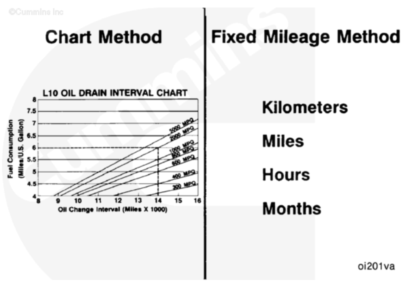
- Fuel Injectors - Replace[3]

1. Mechanically actuated injector engines **only**.
2. Electronically actuated injector engines **only**. Alternative fuel filter change information is available in the fuel filter change interval section of this procedure.
3. Recommended at engine half life to rebuild. Half life varies by application. Contact a Cummins® Authorized Repair Location if the interval can **not** be determined. Alternative injector replacement information is available in the injector replacement interval section of this procedure.
4. Refer to Extended Service Interval Program for Mining Applications, Bulletin 4388840 (/qs3/pubsys2/xml/en/bulletin/4388840.html), and Extended Service Interval Program for Generator Set Applications, Bulletin 5411272 (/qs3/pubsys2/xml/en/bulletin/5411272.html).
5. Lubricating oil change interval can be extended to 2 years, if oil does **not** exceed designated hours, and oil is sampled quarterly with oil analysis parameters remaining within the limits specified in Table 2 of Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060 (/qs3/pubsys2/xml/en/bulletin/4022060.html).

Oil and Oil Filter Change Interval

⚠ **CAUTION** ⚠

The use of a synthetic base oil does not justify extended oil change intervals. Extended oil change intervals can decrease engine life due to factors such as corrosion, deposits, and wear.



⚠ **CAUTION** ⚠

Do not extend oil change intervals beyond the maximum point shown when fuel consumption rates are less than the minimum shown. Doing so can result in shortened engine life due to oil degradation processes associated with lightly loaded engines.

Do **not** extend oil and filter change intervals beyond those specified configurations in the maintenance schedule, unless the chart method is used for industrial applications or Service Bulletin, Extended Lubricating Oil Service Interval Program for Standby Generator Set Applications, Bulletin 6475874 (/qs3/pubsys2/xml/en/bulletin/6475874.html), for standby applications operating under 250 hours. See the oil interval charts in this section.

Centinel™ - With a continuously operating Centinel™ System, the oil drain interval can be extended until the oil analysis requires the oil to be changed or the oil is known to be contaminated. Oil analysis is required at 250 hour intervals when using the Centinel™ System. Oil analysis parameters **must** remaining within the limits specified in Table 2 of the Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060 (/qs3/pubsys2/xml/en/bulletin/4022060.html). Reference the following bulletins:

- Fluids for Cummins® Products Service Manual, Bulletin 5411406, Section 4 - Engine Oil.
- Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060. (/qs3/pubsys2/xml/en/bulletin/4022060.html)
- Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452. (/qs3/pubsys2/xml/en/bulletin/2883452.html)

⚠ CAUTION ⚠

Inaccurate fuel consumption records can result in incorrect oil change intervals that can result in premature engine wear and/or deposits.

⚠ CAUTION ⚠

Do not extend oil change intervals beyond the maximum point shown when fuel consumption rates are less than the minimum shown. Doing so can result in shortened engine life due to oil degradation processes associated with lightly loaded engines.

Oil change intervals can be determined by one of the following recommended methods:

- Fixed-Hour Method (based on fixed hours or months, whichever occurs first)
- Chart Method (based on known fuel consumption rates, coupled with oil analysis program)
- Centinel™ System, coupled with an oil analysis program.
- Lubricating Oil Drain Interval Extension Testing - use the following section in Service Bulletins, Extended Service Interval Program for Mining Applications, Bulletin 4388840 (/qs3/pubsys2/xml/en/bulletin/4388840.html), Extended Service Interval Program for Generator Applications, Bulletin 5411272 (/qs3/pubsys2/xml/en/bulletin/5411272.html), and Extended Service Interval Program for Generator Applications, Bulletin 5659817 (/qs3/pubsys2/xml/en/bulletin/5659817.html). Refer to Section 6.

The information listed below is required when using the chart method to determine the correct oil and filter change interval for the engine.

- Oil system capacity (sump plus any remote tank volume)
- Average fuel consumption rate
- Use engine oil that complied to the standards listed in Fluids for Cummins® Products Service Manual, Bulletin 5411406, Section 4 - Engine Oil.
- Oil analysis to demonstrate the chart method interval is acceptable. Reference the following bulletins:
    - Fluids for Cummins® Products Service Manual, Bulletin 5411406, Section 4 - Engine Oil.
    - Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060. (/qs3/pubsys2/xml/en/bulletin/4022060.html)
    - Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452. (/qs3/pubsys2/xml/en/bulletin/2883452.html)

System capacity can be determined by knowing the volume of the oil required to touch the high-level mark on the dipstick and volume of any remote oil tanks on the machine in which oil is continuously circulated. Oil sump capacities are listed in the operation and maintenance manuals for all Cummins® engines. It is imperative that the total system volume be known when using the chart method. Use the following procedure for information on the oil sump capacity. Refer to Procedure 018-017 in Section V. (/qs3/pubsys2/xml/en/procedures/56/56-018-017.html) If the machine is equipped with an oil reserve system with a reservoir remote from the engine oil sump, the reservoir volume **must** be added to the engine sump volume to determine the total system capacity. This is **only** true for remote tanks in which the oil is continuously circulated. The Centinel™ make-up tank volume does **not** add to system capacity since the engine oil is **not** continuously circulated through this tank.

When oil is changed on an engine with a remote oil tank in which oil is continuously circulated (**not** a Centinel™ make-up tank), the oil in the tank **must** be changed in addition to the oil in the engine sump.

To use the chart method effectively, accurate fuel consumption records **must** be kept and maintained. Fuel consumption information **must** be in liters per hour or gallons per hour units to use the charts. Fuel consumption rates can change over time because of an increasing or decreasing engine duty cycle. Accurate records are essential in determining average fuel consumption during a given oil change interval.

The charts are categorized and labeled by the system capacity for a specific engine family. Select the chart representing the correct oil system capacity for engine model in question. The left vertical axis of the chart represents fuel consumption in U.S. gallons per hour. Determine the intersection point of the fuel consumption rate on the applicable interval curve by drawing a horizontal line. From this point, draw a vertical line down until it intersects with the horizontal axis representing hours. This point represents the acceptable oil change interval.

Note : Before using the charts below it is important to note the grade of oil being used. Reference Table 1. This can affect which chart can be used.

<table>
<tr><th colspan="4">Table 1: Engine Oil Grades</th></tr>
<tr><th>North American Classification</th><th>International Classification</th><th>Cummins® Engineering Standard (CES)</th><th>Description</th></tr>
<tr><td>API[1] CD API CE</td><td rowspan="2">ACEA[2] E1</td><td rowspan="2">Standard Obsolete. These oils are no longer recommend for use in Cummins® engines.</td><td rowspan="4">Do not use the Chart Method if using these oils.</td></tr>
<tr><td>API[1] CG-4/SH</td></tr>
<tr><td rowspan="2">API CF-4/SG</td><td>ACEA[2] E2</td><td rowspan="2">CES 20075[7]</td></tr>
<tr><td>ACEA[2] E3</td></tr>
<tr><td rowspan="3">API[1] CH-4[4] /SJ</td><td>JAMA[3] DH-1</td><td>CES 20071</td><td>Engines using this oil grade must be considered a high load factor engine and must use that line on the relevant chart</td></tr>
<tr><td rowspan="2">ACEA[2] E5[5]</td><td>CES 20076</td><td rowspan="6">Engines using these oil grades can use either line on the charts depending on load factor</td></tr>
<tr><td>CES 20077</td></tr>
<tr><td>API[1] CI-4</td><td>ACEA[2] E7</td><td>CES 20078</td></tr>
<tr><td>API[1] CI-4</td><td>ACEA[2] E7</td><td>CES 20078</td></tr>
<tr><td rowspan="2">API[1] CJ-4</td><td>ACEA[2] E9</td><td rowspan="2">CES 20081</td></tr>
<tr><td>JAMA[3] DH-2</td></tr>
</table>

Table 1: Engine Oil Grades			
North American Classification	**International Classification**	**Cummins® Engineering Standard (CES)**	**Description**

1. API - American Petroleum Institute.
2. ACEA - Association des Constructeurs European d' Association. ACEA - Association des Constructeurs European d' Association.
3. JAMA - Japanese Automobile Manufacturers Association. JAMA - Japanese Automobile Manufacturers Association.
4. CES 20076 adds the requirement of a 300 hour Cummins® M11 test to API CH-4. CES 20076 adds the requirement of a 300 hour Cummins® M11 test to API CH-4.
5. CES 20077 adds the requirement of a 300 hour test to ACEA E5. CES 20077 adds the requirement of a 300 hour test to ACEA E5.
6. Use of oils with **only** these designations poses an undue risk of engine damage for engines designed to use more advanced oils, even when drastically shortened oil change intervals are followed.
7. CES 20075 CF-4/SH and E-3 oils can be used in areas where none of the recommended oils are available, but the oil drain interval **must** be reduced. Reference the appropriate maintenance schedule. Use of oils with **only** these designations poses an undue risk of engine damage for engines designed to use more advanced oils, even when drastically shortened oil change intervals are followed.

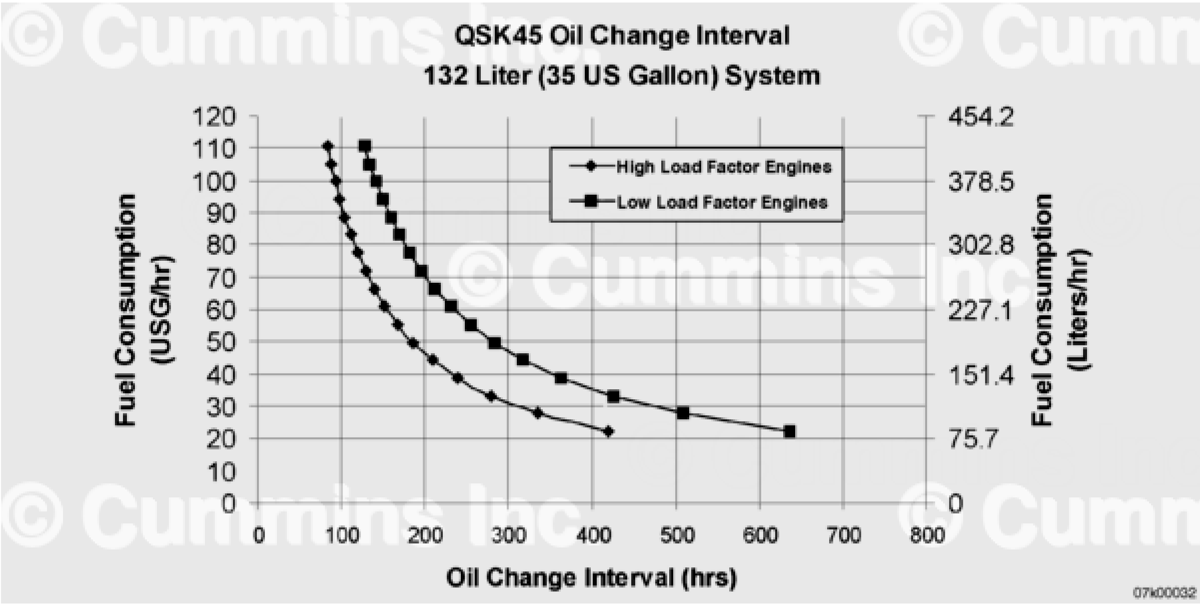
⚠ CAUTION ⚠

Lubricating oils must be classified by one of the standards listed in Table 1. Oils that have never been tested nor received a certificate of compliance from Cummins Inc. cannot be recommended for use in Cummins® products.

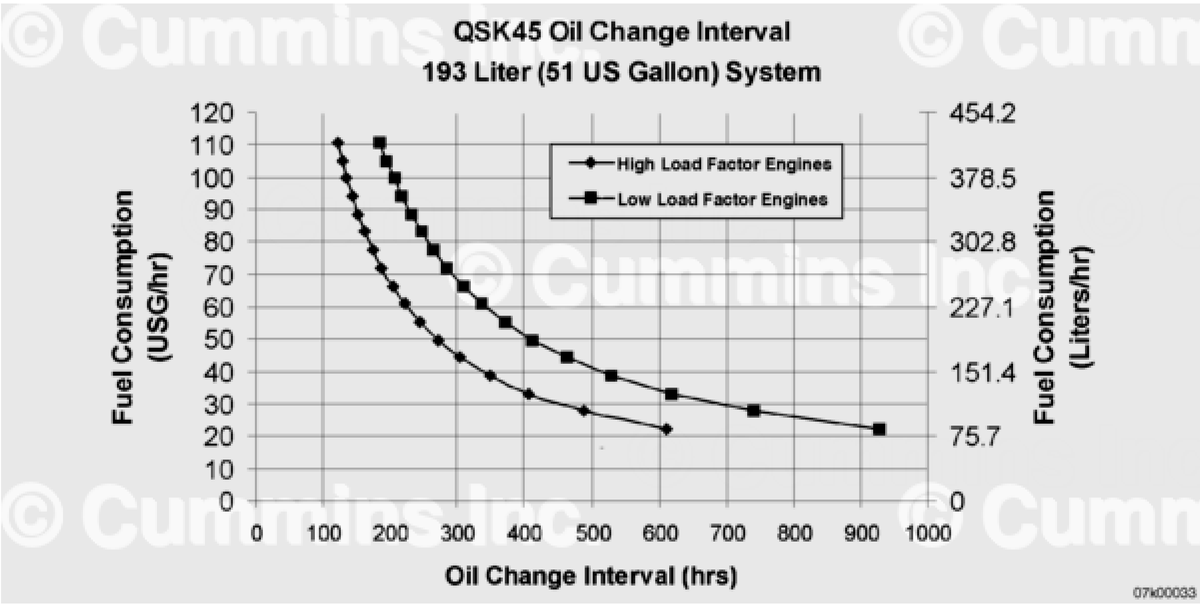
Lubricating Oil Filters

Premium filters are recommended. Premium filters contain synthetic media materials to provide more efficient filtration for the entire service life and to extend the media life over conventional cellulose media. Premium full-flow filters are made with synthetic media and have the StrataPore™ designation on the outside of the filter. StrataPore™ filters have the efficiency, capacity, and strength needed for this extended service.

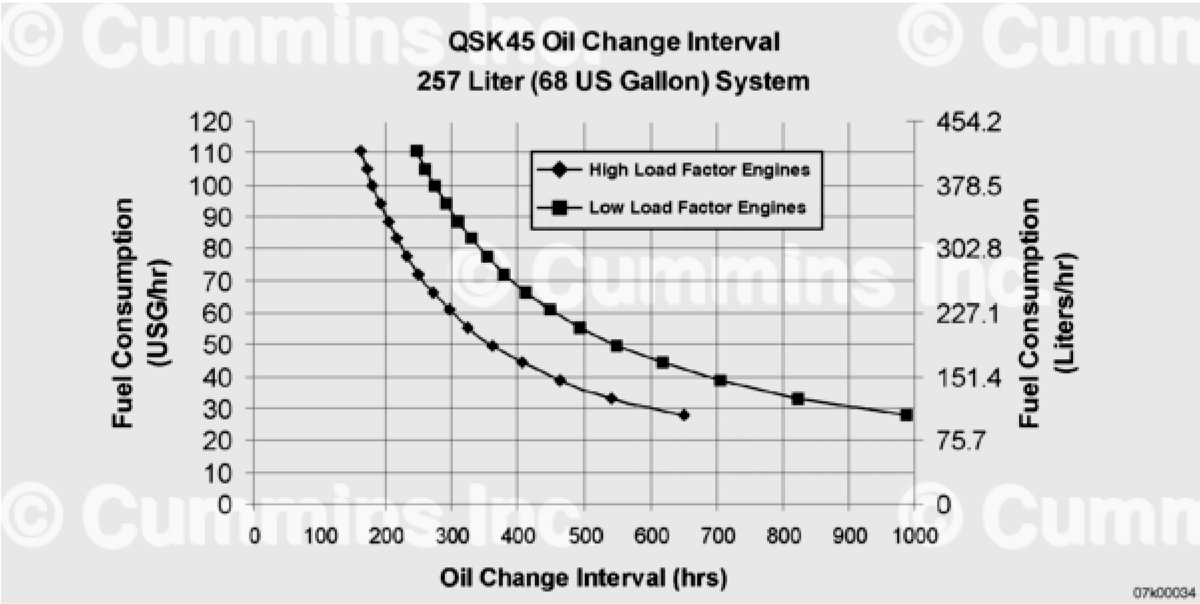
Note : When using the graphs below to calculate oil change interval, High Load Factor engines refers to engines with a load factor over 45%, generally excavators and wheel loaders. Low Load Factor engines are referred to engines with a load factor below 45%. Contact your Local Cummins® Distributor for help calculating engine load factor.



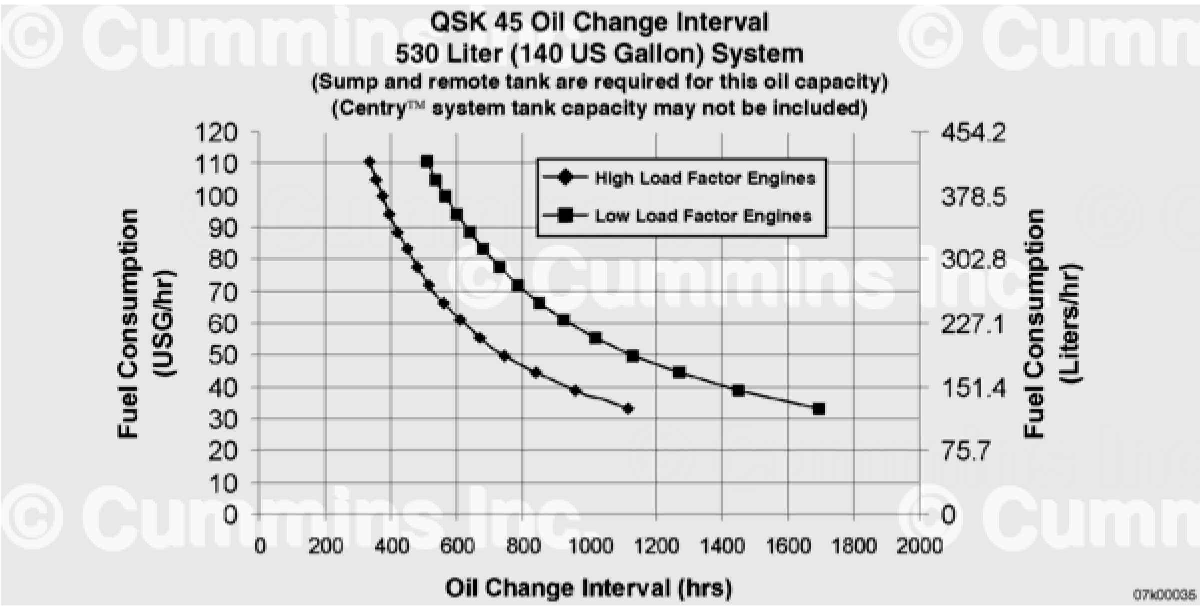
QSK45 Oil Change Interval, 132 Liter (35 Gallon) System



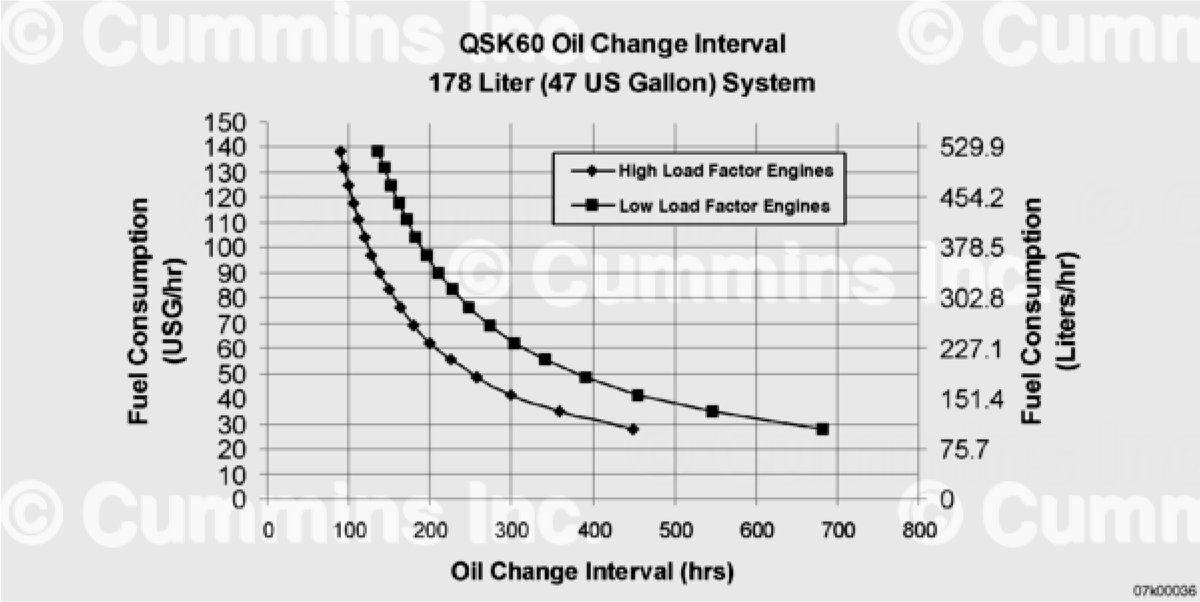
QSK45 Oil Change Interval, 193 Liter (51 Gallon) System



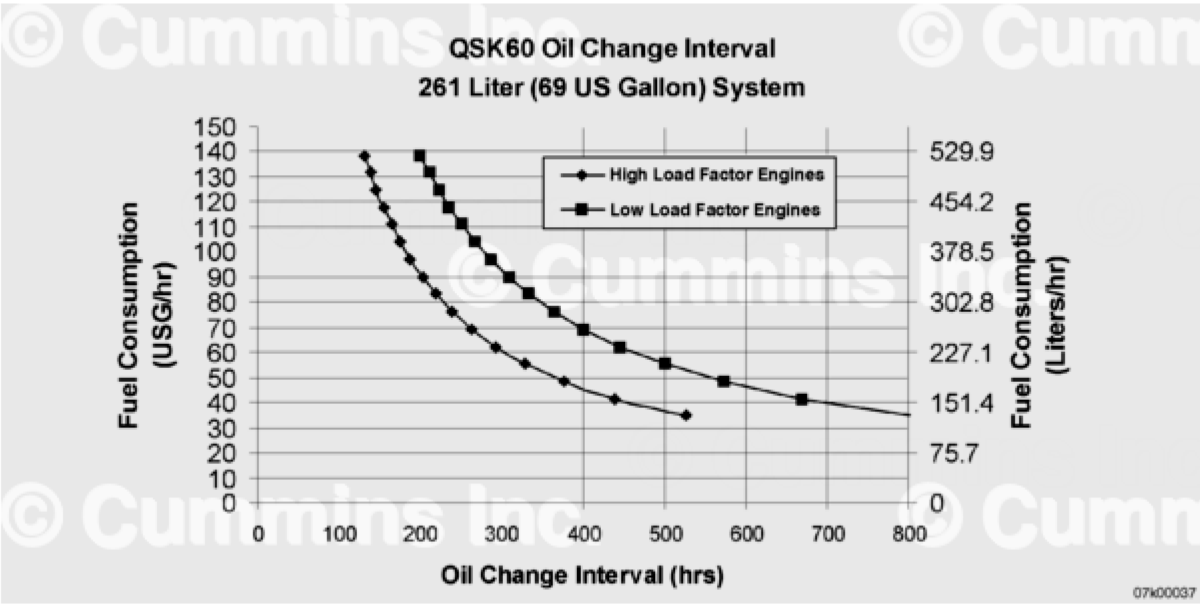
QSK45 Oil Change Interval, 257 Liter (68 Gallon) System



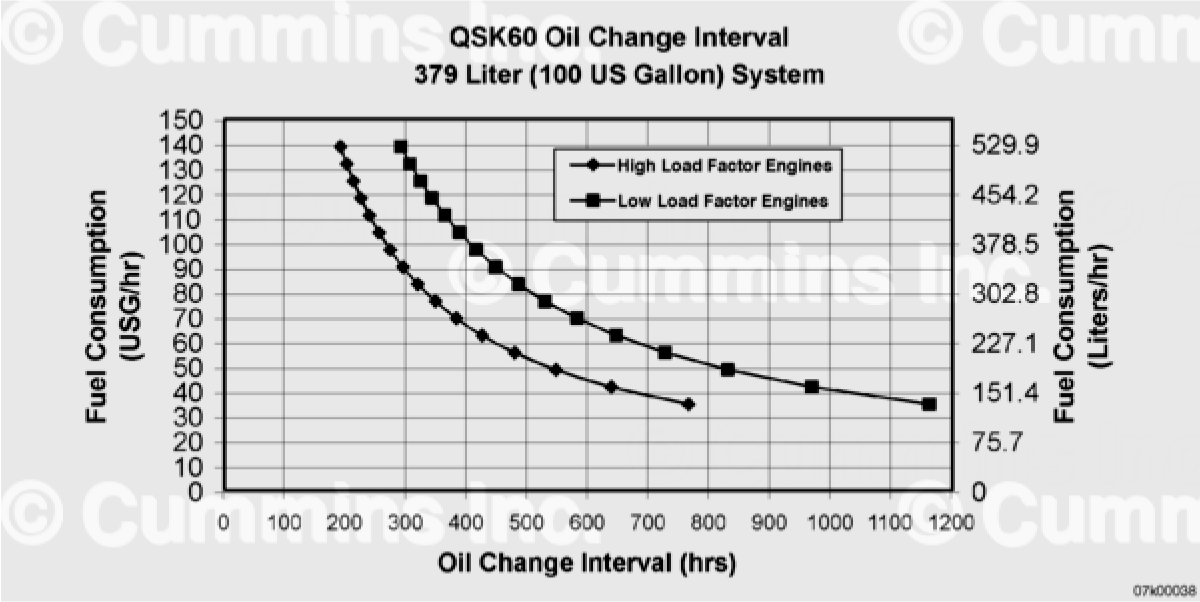
QSK45 Oil Change Interval, 530 Liter (140 Gallon) System



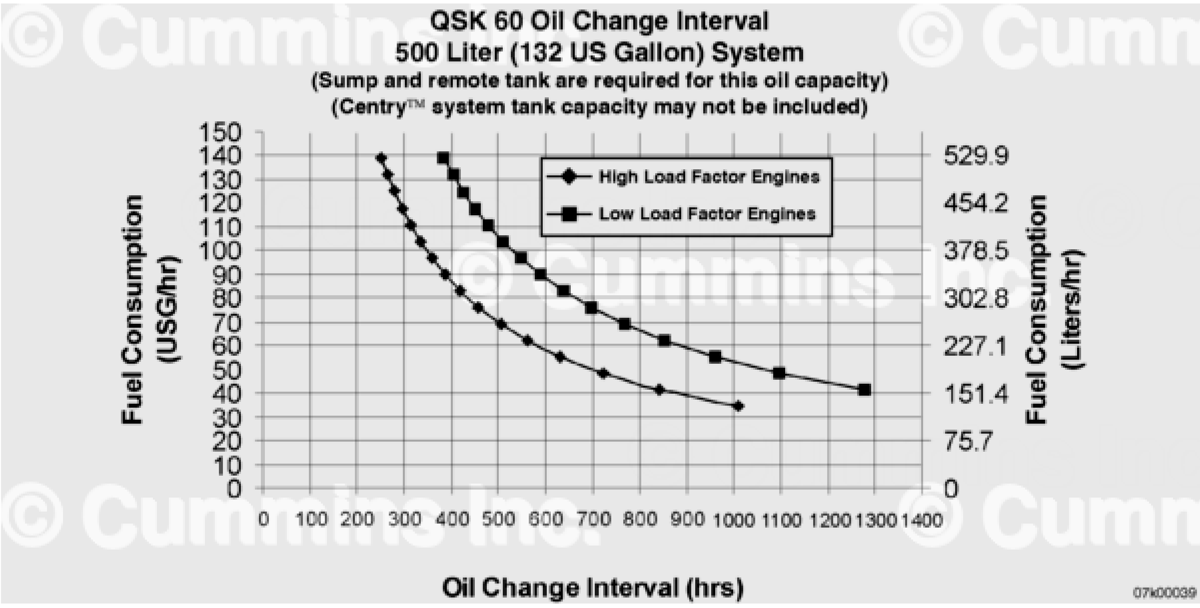
QSK60 Oil Change Interval, 178 Liter (47 Gallon) System



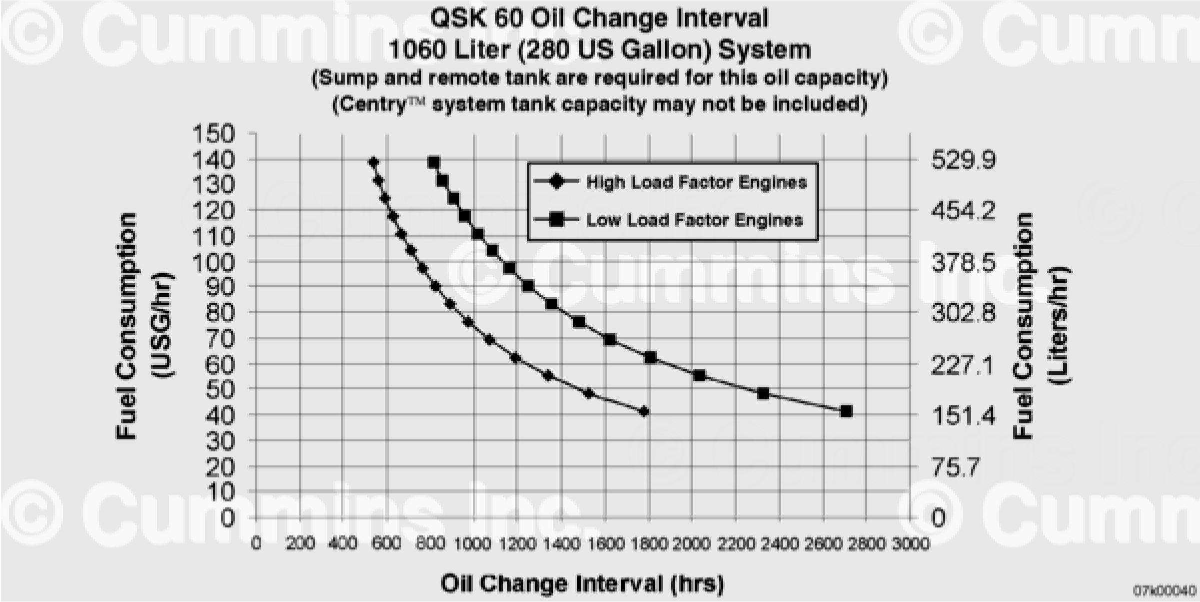
QSK60 Oil Change Interval, 261 Liter (69 Gallon) System



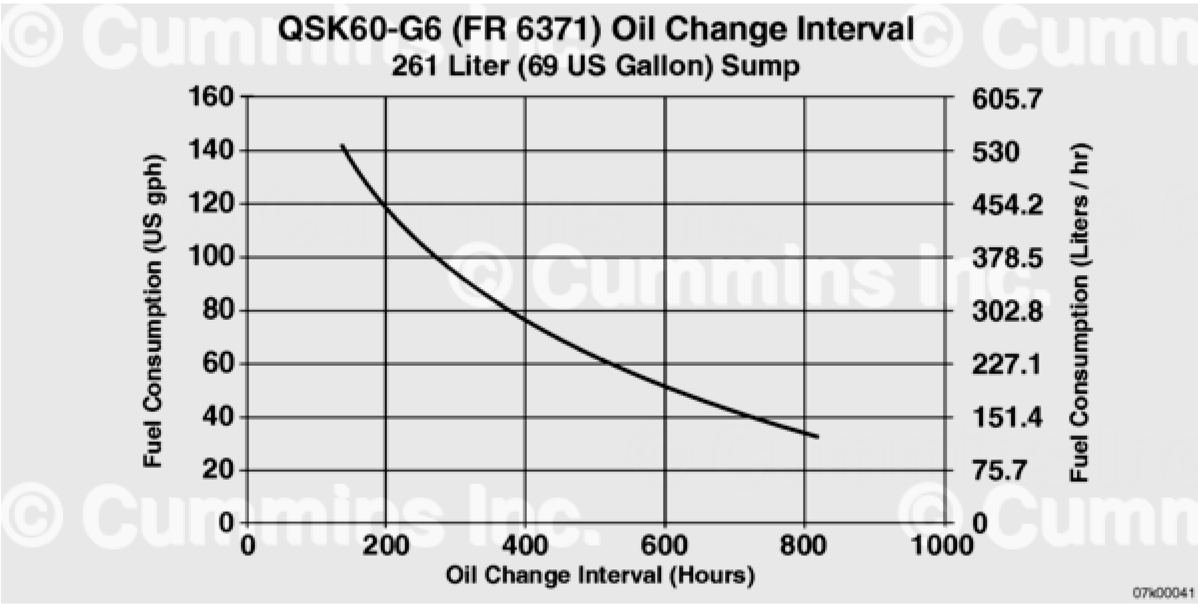
QSK60 Oil Change Interval, 379 Liter (100 Gallon) System



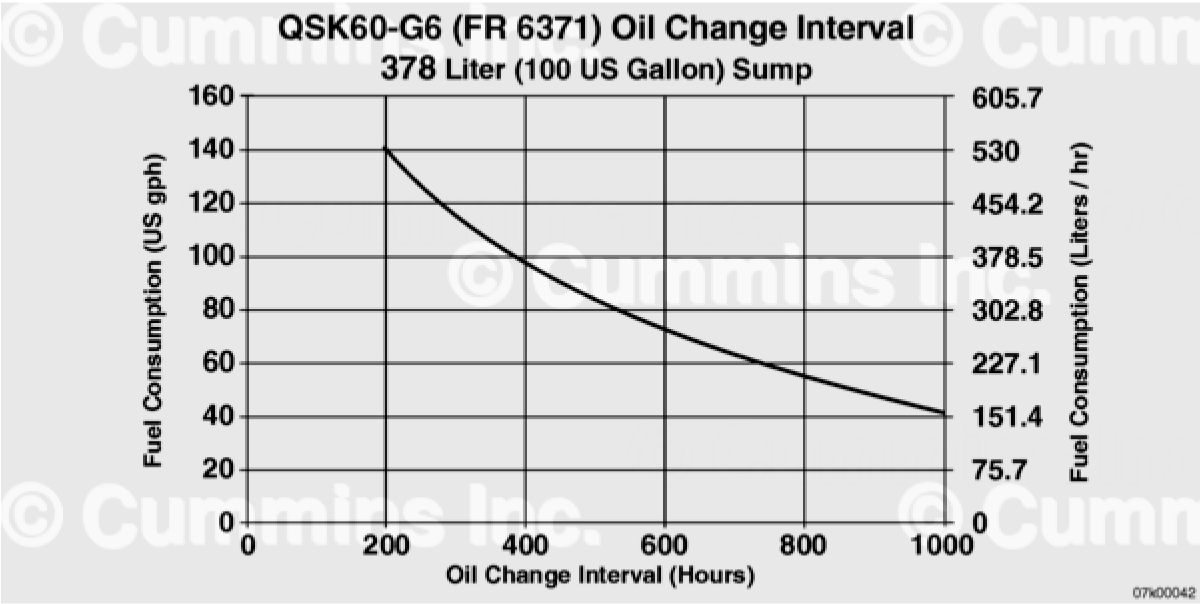
QSK60 Oil Change Interval, 500 Liter (132 Gallon) System



QSK60 Oil Change Interval, 1060 Liter (280 Gallon) System



QSK60-G6 (FR 6371) Oil Change Interval, 261 Liter (69 Gallon) System



QSK60-G6 (FR 6371) Oil Change Interval, 678 Liter (100 US Gallon) System

For oil system volumes **not** shown on the charts, the appropriate oil change interval can be calculated by interpolation between oil change intervals obtained from charts of system volumes below and above the desired system volume. Method:

Variables

- Oil system volume for change interval desired = VS _____
- Oil system volume for chart below desired volume = VB _____
- Oil system volume for chart above desired volume = VA _____
- Oil change interval from chart for VB volume = HRB _____
- Oil change interval from chart for VA volume = HRA _____

1. Read the oil change intervals HRB and HRA from the charts for system volumes below (VB) and above (VA) the desired system volume VS using the fuel consumption level and type of oil used.
2. Calculate the volume ratio factor = VR = (VS - VB)/(VA - VB) (_____ - _____)/(_____ - _____) =
3. Calculate desired oil change interval = HRS = HRB + ((HRA - HRB) x VR) [HRS = _____ + ((_____ - _____) x _____)]

Centinel™ - With a continuously operating Centinel™ System, the oil drain interval can be extended until the oil analysis requires the oil to be changed or the oil is known to be contaminated. Oil analysis is required at 250 hour intervals when using the Centinel™ System and **must** include soot measurement. Fleetguard® Oil Analysis Kit, Part Number CC 2543, meets this requirement. Reference the following bulletins for more information on oil sampling and analysis.

- Fluids for Cummins® Products Service Manual, Bulletin 5411406, Section 4 - Engine Oil.
- Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060. (/qs3/pubsys2/xml/en/bulletin/4022060.html)
- Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452. (/qs3/pubsys2/xml/en/bulletin/2883452.html)

Premium oil filters, such as Fleetguard® LF 9000 series filters, or equivalent, are also required when using Centinel™ with spin-on filters.

Fuel Filter Change Interval

Fuel Filter Change Intervals for Electronically Actuated Injector Engines Only

- Fuel filter change intervals are set at 250 hours in order to protect the fuel system from hard particles.
- **Not** all fuel filters available on the market are equivalent, and may **not** protect the fuel system adequately beyond the 250 hour threshold.
- Cummins Inc. requires the following hardware to achieve Stage 1 and Stage 2 fuel filter change intervals beyond 250 hours, and up to but **not** exceeding 500 hours:
    - FH239 series Industrial Pro or FH240 series Sea PRO® filter head **must** be installed as the Stage 1 fuel filter head.
    - Fleetguard® NanoNet™ media fuel filters **must** be used on Stage 1 and Stage 2 fuel filters.
    - Use the following procedure for fuel filter part numbers. Refer to Procedure 018-024 in Section V. (/qs3/pubsys2/xml/en/procedures/56/56-018-024.html)

Cummins Inc. does **not** recommend fuel filter change intervals exceed 500 hours unless following the requirements in the extended service interval program service bulletins. Refer to Extended Service Interval Program for Mining Applications, Bulletin 4388840 (/qs3/pubsys2/xml/en/bulletin/4388840.html), and Extended Service Interval Program for Generator Set Applications, Bulletin 5411272 (/qs3/pubsys2/xml/en/bulletin/5411272.html).

The most common cause of degraded engine performance is poor fuel quality. Fleetguard® NanoNet™ media filters are designed to function with greater filtration efficiency than other filters, and thus retain more contaminants. If premature filter plugging is observed, additional filtration or improved fuel quality may be necessary to reach desired maintenance intervals. For more information on fuel cleanliness recommendations, reference the "Fuel Cleanliness" section of the Fuels For Cummins® Engines, Bulletin 3379001 (/qs3/pubsys2/xml/en/bulletin/3379001.html).

Injector Replacement Interval

Injector Replacement Interval for Electronically Actuated Injector Engines Only

- Injector replacement intervals are recommended at engine half life to rebuild for some injectors part numbers. For some injector part numbers, if additional requirements are met, the injector replacement interval is at rebuild.

Injector Replacement Interval		
Injector Part Number	**Description**	**Replacement Interval**
4924589, 4928105, 4955259	C1 Injectors	Half life to rebuild
4964173, 2881085	C2 Injectors	Half life to rebuild
2881089	C3 Injectors without flow limiters	Half life to rebuild
2867148	C3 Injectors with flow limiters, not using Nanonet fuel filters[1] or with fuel sulfur levels > 500 ppm	Half life to rebuild
2867148	C3 Injectors with flow limiters, using Nanonet fuel filters[1], fuel sulfur levels <500 ppm	Rebuild

[1]Nanonet Fuel Filtration:

- FH239 series Industrial Pro or FH240 series Sea Pro® filter head **must** be installed as the Stage 1 fuel filter head.
- Fleetguard® NanoNet™ media fuel filters **must** be used on Stage 1 and Stage 2 fuel filters.
- Use the following procedure for fuel filter part numbers. Refer to Procedure 018-024 (/qs3/pubsys2/xml/en/procedures/56/56-018-024.html) in Section V or Extended Service Interval Program for Mining Applications, Bulletin 4388840.

Coolant Drain Interval

If following the requirements in the extended service interval program service bulletins, coolant **only** needs to be replaced if indicated through fluid sampling. Refer to Extended Service Interval Program for Mining Applications, Bulletin 4388840 (/qs3/pubsys2/xml/en/bulletin/4388840.html), and Extended Service Interval Program for Generator Set Applications, Bulletin 5411272 (/qs3/pubsys2/xml/en/bulletin/5411272.html).

Maintenance Intervals for Cooling System up to 568 liters [150 gal]					
	System Size in liters [gal]				
	79-144 [21-30]	117-189 [31-50]	193-284 [51-75]	288-378 [76-100]	382-568 [101-150]
Hours	Units of SCA				
751-1000	25	50	80	75	150

Maintenance Intervals for Cooling System up to 568 liters [150 gal]					
	System Size in liters [gal]				
501-750	20	35	60	100	110
251-500	15	25	40	50	75
0-250	10	15	20	25	40

Maintenance Intervals for Cooling Systems up to 1514 liters [400 gal]					
	System Size in liters [gal]				
	572-757 [151-200]	761-946 [201-250]	950-1135 [251-300]	1139-1325 [301-350]	1329-1514 [351-400]
Hours	Units of SCA				
751-1000	200	250	300	350	400
751-1000	150	190	225	260	300
251-500	100	125	150	175	200
0-250	50	65	75	90	100

A. Consult the vehicle equipment manufacturer's maintenance information for total cooling system capacity.
B. When draining and replacing the coolant, **always** pre-charge the cooling system to a SCA level of 1.5 units per gallon. This concentration level **must not** be allowed to go below 1.2 units and **must** be controlled when the level is greater than 3 units. Action needed when the level goes below 1.2 is a filter and liquid pre-charge; from 1.2 to 3.0 units, filter **only**; above 3.0, test at every oil change until the level falls to 3.0 or below.
C. Use the following procedure for coolant filter part numbers and SCA capacity. Refer to Procedure 018-024 in Section V. (/qs3/pubsys2/xml/en/procedures/56/56-018-024.html)

When performing service which requires draining the cooling system, take special precautions to collect it in a clean container, seal it to prevent contamination, and save for reuse.

A. Change coolant filters at each oil change to protect the cooling system. Consult the coolant capacity chart to determine the correct coolant filter for a given cooling system capacity and oil drain interval.